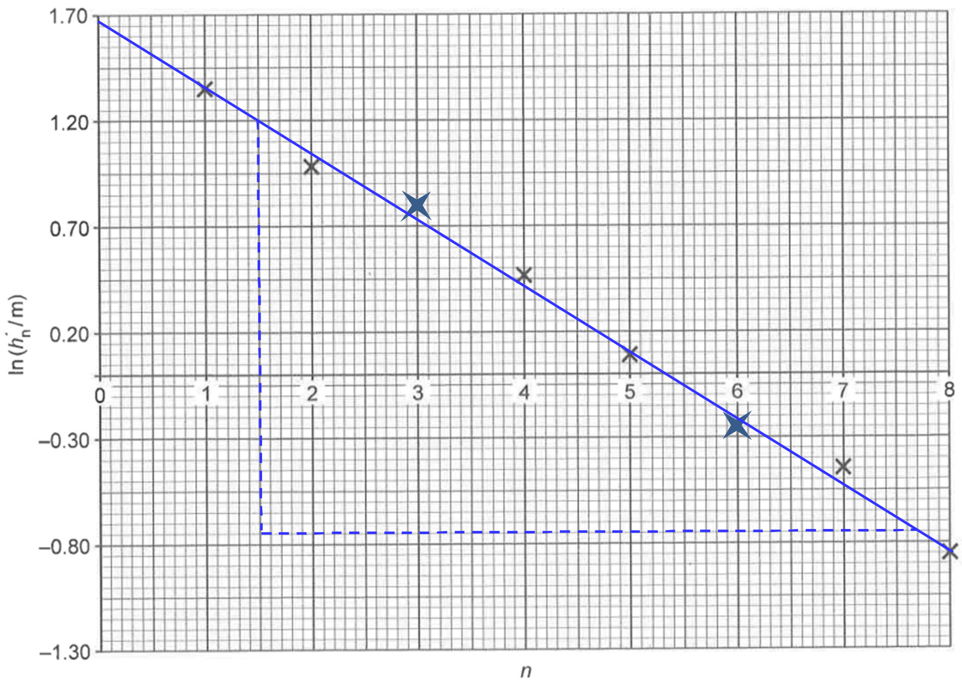
4(a)(i)	Gravitational potential energy converts to kinetic energy.
4(a)(ii)	For downward journey, Gain in KE = Loss in GPE Final KE $=(0.430)(9.81)(3.40)=14.3\ \text{J}$
4(a)(iii)	For upward journey, Loss in KE = Gain in GPE Initial KE $=(0.430)(9.81)(2.95)=12.4\ \text{J}$
4(a)(iv)	$\frac{h_1}{h_0}=\frac{2.95}{3.40}=0.86765$ $\frac{h_2}{h_1}=\frac{2.36}{2.95}=0.80000$ $\frac{h_3}{h_2}=\frac{1.75}{2.36}=0.74153$

No.	Solution
	$\frac{h_4}{h_3} = \frac{1.21}{1.75} = 0.69143$ $\frac{h_5}{h_4} = \frac{0.78}{1.21} = 0.64463$ The ratio decreases. Therefore, the student is not correct.
4(b)(i)	$n = 3,\ \ln h_n = 0.80$ $n = 6,\ \ln h_n = -0.25$
4(b)(ii)	 **Fig. 4.1**
4(c)(i)	The line of best fit is a straight line with a negative gradient. Plotting $\ln h_n$ against n yields a straight line with a gradient $= -k$, and k is a constant.
4(c)(ii)	Gradient = $\frac{-0.750 - 1.200}{7.70 - 1.50} \approx -0.31$ Therefore $k = 0.31$
4(c)(iii)	The vertical intercept of the graph is $\ln h_0$. Therefore the initial height h_0 is obtained using $h_0 = e^{\text{vertical intercept}}$.
4(c)(iv)	From graph, $\ln h_0 = 1.675$ Initial height $h_0 = e^{1.675} \approx 5.3$ m

No.	Solution
5	Nuclear fusion is the joining of two smaller nuclei to form a larger nucleus